How are theories created?

1 Use the words below to fill the gaps in these sentences.
You may need to use a word more than once.

theory	mixture	experiments	thinker	atoms

Parmenides was a Greek philosopher or _____ who lived 2500 years ago. He had the idea that everything was made of one kind of stuff. Other thinkers said that everything was made up of earth, air, fire or water or a _____ of these.

Leusippus and Democritus developed a different _____, that everything was made up of different kinds of tiny particles that they called _____, but their theory was forgotten for 1800 years.

Many _____ have shown that everything is made up of atoms, so they were right all the time!

2 Join each box on the left to one or more boxes on the right to make short sentences about solids, liquids and gases.

Solids	**change size**
Liquids	**change shape**
Gases	**keep the same size**
	keep the same shape

ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

How do solids, liquids and gases behave?

1 Use the words below to fill the gaps in these sentences. You will need to use some words more than once and other words not at all.

spread out	can	cannot	faster	slower	expand	contract

Solids _____ be squashed.

Liquids _____ be squashed.

Gases _____ be squashed.

When substances are heated, they usually _____. And whenever they can, substances _____ _____.

Usually, the hotter something is, the _____ it spreads out.

2 For each of the substances below, say whether it is a solid, a liquid or a gas by joining it to the right box.

Paper clip	
Iron nail	solid
Air	
Oxygen	
Carbon dioxide	liquid
Petrol	
Milk	
Paper	gas
Beer	

ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

Differences between solids, liquids and gases

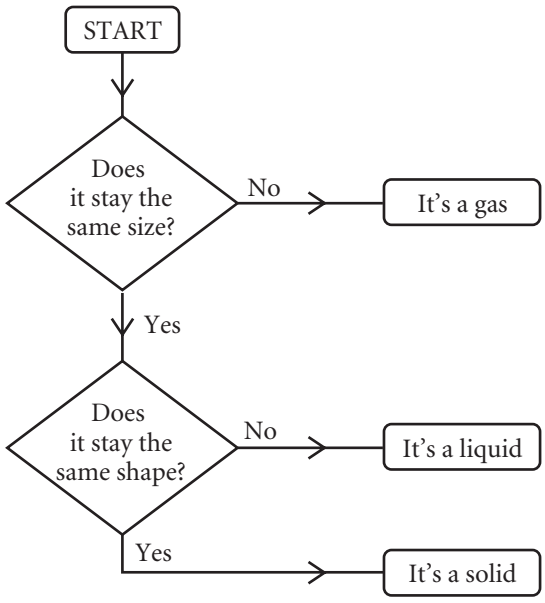
1 Use the key below to fill the gaps in these sentences. You may need to use a word more than once.

quickly	solids	gas	mixture	flows	solid

Sometimes it is not easy to decide whether a substance is a solid, a liquid or a gas. This usually happens when the substance is a _____. For example, tomato sauce is a mixture of _____ and liquids, so it behaves a bit like a liquid (it _____) and a bit like a solid (it doesn't flow very _____ and it keeps its shape a bit).

Other mixtures might be a gel, which is a mixture of a _____ and a liquid, or a foam which can be a mixture of a _____ and a solid or a mixture of a _____ and a liquid.

2 Use the key opposite to say whether substances A, B and C are either solids, liquids or gases.



Substance A has a volume of 200 cm^3, which never changes, but it fills the bottom of the beaker it is poured into. Substance A is a _____.

Substance B can change its size and shape depending on the container it is in. Substance B is a _____.

Substance C can only change its size or shape when it is cut up, for example with a saw. Substance C is a _____

℗ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

The particle model of solids, liquids and gases

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

cannot	particle	spread	solid	small	close	slide

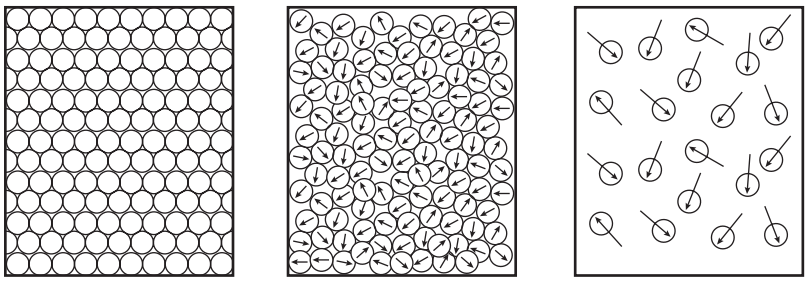
A _____ is a tiny bit of something. The particles in substances are so _____ that they can hardly be seen even with a microscope.

In a solid, the particles are _____ together and they _____ move around.

In a liquid the particles are still as close together as they are in a _____, but they are not fixed in place so they can _____ past each other.

In a gas the particles are _____ out and are moving around very quickly in all directions.

2 Label these diagrams either ''solid'', ''liquid'' or ''gas'' to show which diagram shows the arrangement of particles for that state of matter.



ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme

Does the particle model explain anything else?

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

higher	walls	pressure	removed	particles	diffusion	outside

Gas _____ push against the walls of whatever they are in because they are moving around and hit the walls of the container . This is called _____. The more gas particles hit the _____ each second, the _____ the pressure.

When the air is _____ from a can, the can collapses because air particles are pushing from the _____.

Because particles move about, they tend to spread out. The scientific name for this process of spreading out is _____.

2 True or false? Look at the statements below. If they are true, write ‘‘T’’ in the box, if they are false, write ‘‘F’’ in the box.

Statements	**True/false**
Particles can expand.	☐
Gas pressure is caused by moving particles.	☐
Diffusion is caused by moving particles.	☐
An empty can is full of air.	☐
A can collapses when the air is sucked out because it is sucked inwards.	☐
A can collapses when the air is sucked out because it is pushed inwards by air pressure.	☐
Dust motes in the air move because air particles hit them.	☐

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How can we tell when a liquid is a mixture?

1 Use the words below to fill the gaps in these sentences. You may need to use a word more than once.

dissolve	mixtures	soluble	insoluble	solutions	filtering

If you take sugar or salt and _____ it in water you can't see it any more.

Sugar is _____ which means that it will dissolve. Sand is _____ which means that it won't dissolve.

Solutions are _____, because you can get back the substance you dissolved by heating the solution until the liquid has evaporated leaving the substance behind.

If a substance in a mixture is insoluble you can sometimes separate it by _____.

2 Complete the table.

Solid	Liquid	Does it dissolve?	Name of solute
salt	water	yes	
sugar	water	yes	
copper sulphate	water	yes	
sand	water	no	

ɑ I Bradley, C Tear, M Winterbottom, S Young, 2001, The Heinemann Science Scheme